

A Cyberpunk 2077 perspective on the prediction and understanding of future technology☆

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Highlights

Cyberpunk 2077 offers insights into AI, edge computing, biotech, and augmented humans. Key developments include BCIs, multimodal systems, VR, and AI-driven appliances. There is a growing need for technologies compatible with existing infrastructure. Sci-fi games like Cyberpunk 2077 help shape public views on future innovations.

Abstract

Science fiction and video games have long served as valuable tools for envisioning and inspiring future technological advancements. This position paper investigates the potential of Cyberpunk 2077, a popular science fiction video game, to shed light on the future of technology, particularly in the areas of artificial intelligence, edge computing, augmented humans, and biotechnology. By analyzing the games portrayal of these technologies and their implications, we aim to understand the possibilities and challenges that lie ahead. We discuss key themes such as neural links and brain-computer interfaces, multimodal recording systems, virtual and simulated reality, digital representation of the physical world, augmented and AI-based home appliances, smart clothing, and autonomous vehicles. The paper highlights the importance of designing technologies that can coexist with existing preferences and systems, considering the uneven adoption of new technologies. Through this exploration, we emphasize the potential of science fiction and video games like Cyberpunk 2077 as tools for guiding future technological advancements and shaping public perception of emerging innovations.

Keywords

Future technology
Videogames
Cyberpunk
Predictions

1. Introduction

The rapid pace of technological advancements has led to an ever-increasing interest in understanding and predicting the future of technology. The capacity of science fiction to stimulate actual scientific exploration, technological advancement and their impact on society remains a topic of sustained fascination [1], [2], as seen in works such as Neuromancer [3], Snow Crash [4], Do Androids Dream of Electric Sheep? [5], Little Brother [6], Star Trek [7] and The Three-Body Problem [8]. Film serves as a compelling medium for illustrating the intricate relationship between science fiction and technological progress. These cinematic depictions enrich the ongoing scholarly discourse on the ethical and technological implications of emerging innovations, particularly in sectors like healthcare and aerospace [9]. More recently, video games have emerged as an interactive medium that offers unique opportunities to explore futuristic concepts and ideas, with some experts arguing that these ideas are inevitable [10]. Fig. 1 shows an example.

One such video game, Cyberpunk 2077, has garnered attention for its portrayal of a technologically advanced society and the potential challenges and opportunities it presents. The games developers have detailed the artistic and technological aspects of creating its immersive cyberspace [11]. Its companion book, The World of Cyberpunk 2077 [12], offers an in-depth look into the games lore and setting and provides a comprehensive understanding of the games world.

Despite the rich narratives provided by science fiction literature and film, there is a significant gap in how these narratives connect with real-world technological foresight and societal implications. Current academic literature often focuses on speculative depictions of the future without sufficiently addressing how these visions intersect with contemporary technological trends and societal challenges. This paper addresses this gap by positioning *Cyberpunk 2077* as a form of entertainment, but also a crucial cultural piece that bridges speculative fiction and technological foresight, offering a unique, interactive way to engage with emerging technologies and their potential impact.

Set in a dystopian future, *Cyberpunk 2077* presents a detailed vision of an urban landscape dominated by powerful corporations that shape and govern the direction of technological advancements. This backdrop vividly intertwines social, political, and economic themes, providing a rich narrative for players to navigate. *Cyberpunk 2077* has not only captured the imagination of gamers worldwide but has also sparked discussions among academics, researchers, and industry professionals about the likelihood and implications of such a future [13], [14], [15], [16].

The game presents a nuanced view of technology, exploring themes such as disability, gender, and the relationship between humans and machines [17], [18]. These themes raise significant ethical and social questions, illustrating the potential of video games as a medium for stimulating debate and influencing future advancements. As an illustration, Fox [15] delves into the portrayal of disability in *Cyberpunk 2077* and the potential trajectory of cyborg healthcare. Notably, while the game offers detailed depictions of technologically advanced societies, history underscores that the uptake of innovations is often uneven. This disparity is shaped by factors like economic differences, cultural inclinations, and regulatory landscapes [19], [20], [21], a notion echoed profoundly in the game [12].

Cyberpunk 2077, as a significant work within the science fiction genre, presents a unique and immersive exploration of futuristic technologies and their societal implications. Unlike traditional science fiction narratives, *Cyberpunk 2077* provides players with interactive, playable scenarios that dig into the ethical, social, and moral dilemmas posed by emerging technologies.

Due to this, *Cyberpunk 2077* stands out within the science fiction landscape by providing an interactive, immersive experience that goes beyond traditional narratives. Unlike other media, the game allows players to actively engage with scenarios that reflect ethical, social, and moral dilemmas posed by advanced technologies. Themes such as disability, gender, and the complex relationship between humans and machines are explored in nuanced ways that raise significant ethical and social questions [17], [18]. These themes are reflective of ongoing technological debates and also provide a speculative, but also predictive lens through which we can consider the trajectory of technological innovations.

In our contribution, we examine the key technological themes and advancements presented in *Cyberpunk 2077* and compare them with the current state of technology, aiming to move beyond a mere inventory of these technologies and themes by analyzing how they are integrated into the games narrative and specially world-building, providing unique insights that are both reflective and predictive of real-world technological advancements. By doing so, we identify existing challenges and opportunities for further development in various domains, including human augmentation, brain-computer interfaces, simulated reality, digital representation, smart homes and appliances, autonomous vehicles, and advanced smart assistants. Furthermore, we discuss the importance of designing technologies that can coexist with existing user preferences and technologies, ensuring a gradual and effective transition towards a more technologically advanced society [10], [15].

The novelty of our work lies in its approach to analyzing *Cyberpunk 2077* as a foresight tool that critically examines the integration of advanced technologies within a complex social fabric. In this context, we aim to move the technological foresight frontier forward by systematically comparing the games depiction of technological advancements, such as human augmentation, brain-computer interfaces, and autonomous systems, with current state-of-the-art technologies and research. By

contextualizing Cyberpunk 2077 within the broader landscape of science fiction, we highlight how the game aligns with or diverges from existing technological trends and predictions, offering insights that are both reflective of current advancements and predictive of future possibilities.

2. Literature review

In this section, we review existing literature on the role of science fiction in predicting and inspiring technological advancements, as well as studies on the specific technologies mentioned in the presentation. Our aim is to establish the context and show how the papers ideas fit into the broader academic conversation.

2.1. Intersection of science fiction and technological foresight

The intersection of science fiction and technological foresight has been widely explored in recent studies, revealing how speculative narratives can offer valuable insights into the governance, development, and public perception of emerging technologies. Science fiction, as a genre, not only reflects societal anxieties and aspirations but also serves as a tool for imagining alternative futures that challenge conventional thinking about technological progress.

Recent works have highlighted the role of creative writing techniques from science fiction and fantasy in enhancing technology foresight processes. Marshall et al. introduced the Story Thinking model, which applies skills from science fiction and fantasy writing to help transdisciplinary teams better inhabit possible futures, empathize with affected stakeholders, envision plausible scenarios, and engage in shared visions of technology's impact [22]. This model complements existing foresight approaches by enabling researchers to explore complex dimensions of large-scale technological projects in ways that traditional methods often overlook. Similarly, Kymalainen introduced science fiction prototyping as a method for discussing socio-technical issues within emerging technology research [23]. This approach allows researchers to experiment with speculative scenarios, using storytelling as a medium to anticipate the socio-ethical implications of technological innovations. By integrating narrative techniques into foresight practices, these methodologies enhance the capacity to explore diverse and often unforeseen outcomes of new technologies.

The theoretical grounding of futures and foresight studies has been a persistent challenge, as highlighted by Münch et al. who conducted a bibliometric review demonstrating the limited application of scientific theories within the field [24]. This lack of a robust theoretical base has hindered the recognition of foresight studies as a rigorous academic discipline. However, integrating theories from social sciences, such as critical theory, can enrich the understanding of how technologies shape and are shaped by societal forces. Freedman argued that science fiction is a privileged genre for critical theory, serving as a fertile ground for examining the socio-political and ethical dimensions of technological change [25]. By tracing the relationships between science fiction and critical theory, Freedman illustrated how speculative narratives can provide a nuanced critique of contemporary socio-technical systems. This perspective is essential for understanding the broader implications of emerging technologies and for developing foresight practices that are critically aware of underlying power dynamics.

Science fiction has a profound influence on public perceptions and expectations of technology, shaping both the development of new innovations and their acceptance by society. Wilson's study on the impact of science fiction futures depicted in popular culture found that these narratives directly affect how the public perceives and interacts with new technologies, particularly in fields like robotics and human-computer interaction [26]. By reflecting technological dreams and nightmares, science fiction helps frame the discourse around emerging technologies, influencing both public opinion and policy-making. Miller argued that science fiction-inspired approaches offer significant contributions to the governance of new technologies by enhancing the capacity to reflexively assess the social dynamics of socio-technical systems [27]. These approaches allow for a more nuanced understanding of how technological developments might unfold in different contexts, enabling better-informed decisions that consider both technical feasibility and societal impact.

Incorporating decolonial perspectives into foresight practices can provide a more just and ethical approach to technology development. Mohamed et al. explored the application of decolonial theory as sociotechnical foresight in artificial intelligence, proposing tactics to align AI research with principles of justice and equity [28]. This approach emphasizes the importance of considering diverse cultural and ethical perspectives when envisioning the future of technology, ensuring that technological advancements contribute to greater societal well-being rather than perpetuating existing inequalities. Schicks examination of speculative bioethics further highlights the need to critically assess visions of future technologies [29]. As speculative technologies increasingly shape present-day ethical discussions, there is a growing imperative to ground these debates in the realities of current technological capabilities and societal needs.

Design fiction has emerged as a powerful tool for engaging the public in discussions about the potential outcomes of new and emerging technologies. Heidingsfelder et al. developed a design fiction methodology that encourages public reflection on the ramifications of technological advances, bridging the gap between speculative futures and real-world concerns [30]. By creating narrative-driven, speculative artifacts, design fiction fosters critical engagement with possible technological trajectories, making the abstractions more tangible and the future more accessible. However, while these studies explore various aspects of science fictions impact on foresight, they often lack an interactive component that engages users directly with the speculative technologies. This paper addresses this gap by examining *Cyberpunk 2077* as an interactive medium that not only depicts futuristic technologies but also allows players to actively engage with the ethical and societal dilemmas these technologies present. By situating *Cyberpunk 2077* within the context of science fiction and foresight, we can offer a novel perspective on how interactive narratives can enhance our understanding of technological futures.

2.2. Science fiction and technological advancements

Science fiction has a long history of providing a platform for speculating about future technological innovations and their impact on society. The genre often serves as a source of inspiration for researchers and technologists, with numerous real-life inventions tracing their origins to ideas first presented in science fiction stories [31], [32]. The influence of William Gibson is also notable. While his novel *Neuromancer* [3] did not presage the internet itself, which had its origins in the ARPANET of the 1960s and the adoption of TCP/IP in 1983, Gibsons work provided an influential vocabulary – cyberspace – and a vision of a world transformed by networked computing and advanced cyborg technologies. This vision strongly shaped popular discourse about the World Wide Web in the 1990s. Gibsons work also inspired VR researchers, as detailed in Benedikts *Cyberspace: First Steps* and Cavallaros *Cyberpunk & Cyberculture: Science fiction and the work of William Gibson* [33], [34]. The 1995 film *Johnny Mnemonic*, also based on William Gibsons story [35], played a significant role in popularizing virtual reality technologies, fictional depictions of VR are older, dating back to stories such as Vernor Vinges influential *True Names*. Moreover, other films slightly before *Johnny Mnemonic* helped shape public anticipation of VR. For a thorough history of VRs influence on cinema, see David Kirbys *Lab Coats in Hollywood: Science and the Cinema* (MIT Press, 2011) [36]. The 1966 television series *Star Trek* presciently introduced the concept of mobile communication devices, a vision that materialized with Motorolas launch of the inaugural flip phone, the *StarTAC*, in 1996 [37], [38], [39]. Similarly, the idea of a tablet computer can be traced back to Arthur C. Clarks *2001: A Space Odyssey* [40]. Stanley Kubricks seminal 1968 work, provided also an early examination of the ethical and computational complexities inherent in artificial intelligence, exemplified by the character HAL 9000 in the subsequent same named film [41]. The depiction of 3D holographic technology in George Lucass *Star Wars* [42] series has become iconic and inspired practical developments in the field. For a detailed documentation of the history and connection to holography, see Sean F. Johnstons *Holographic Visions: A History of New Science* (Oxford, 2006) [43]. Rajaniemis *The Quantum Thief* [44] explored the transactional nature of data privacy and access rights with the depiction of the *gevolot*, which echoes in the current suggested use of blockchain for

the authentication of information [45], [46]. Furthermore, *Blade Runner 2049* explores the technological feasibility and ethical considerations of 3D printing in the context of creating synthetic humans, a concept inching closer to reality due to advancements by firms such as Organovo [47].

Researchers have explored the symbiotic relationship between science fiction and technological development, arguing that the genre can serve as a catalyst for innovation by providing a vision of potential future technologies [48]. Moreover, science fiction has been used as a tool for technology forecasting, helping to identify emerging trends and assess their potential impact on society [49], [50]. Given its immersive and interactive nature, the cyberpunk genre – especially through mediums like video games – may represent the next frontier in predicting and actively inspiring technological futures, offering insights both ethical, social, and technological dilemmas. Despite these valuable contributions, existing literature often lacks an in-depth examination of how interactive, immersive mediums like video games can serve as powerful tools for foresight and technological speculation. We analyze *Cyberpunk 2077* as a unique interactive narrative that mirrors contemporary technological trends offering a dynamic platform to explore the societal implications of future technologies. The most popular media often lack an interactive component that engages users directly with speculative technologies. There are comparable works to *Cyberpunk 2077* themes in TV-shows such as *Black Mirror* [51] and *Westworld* [52], movies such as *The Matrix* [53]. Even other videogames such as *Deus Ex* [54], and *Detroit: Become Human* [55], also explore themes of technology and society. However, although *Black Mirror* critiques individual technologies, often through standalone narratives that limit exploration to isolated scenarios, and both *The Matrix* and *Westworld* address AI and simulated realities, they rely on linear storytelling that restricts audience interaction and immersion. On contrast, *Deus Ex* and *Detroit: Become Human* share *Cyberpunk 2077*'s focus on human augmentation and AI ethics, but their linear narratives, even when allowing the player to make multiple choices, limit user interaction to somehow predefined pathways. What sets *Cyberpunk 2077* apart is its interactivity set in an open world, allowing players to directly experience and influence these speculative futures, and make their character live in different ways, making it a uniquely immersive foresight tool.

2.3. Video games as a medium for future technologies

As the video game industry has evolved, it has increasingly served as a platform for exploring futuristic concepts and ideas. Video games offer a unique opportunity to engage with potential future technologies in an interactive manner, enabling players to immerse themselves in the game world and experience the consequences of these technologies first-hand [56], [57]. Some recent studies have focused on the potential of video games for inspiring and predicting technological advancements, highlighting the role of game developers in creating plausible and engaging visions of the future [58], [59].

2.4. The unique contributions and predictive potential of *Cyberpunk 2077*

Cyberpunk 2077, while based on the earlier *Cyberpunk* tabletop RPG [60], offers a distinct and immersive exploration of futuristic technologies through its interactive gameplay and richly detailed world-building. Unlike static representations in books or films, the game allows players to experience the implications of these technologies first-hand, creating a unique and impactful narrative experience that can inspire and inform future technological advancements. The interactive nature of *Cyberpunk 2077* provides players with a unique opportunity to engage with and reflect on complex technological and ethical issues. *Cyberpunk 2077*'s detailed world-building sets it apart from other works in the genre. The game's narrative intricately weaves together various technological themes within the socio-political landscape of Night City.

Cyberpunk 2077 has made a significant impact on popular culture, sparking widespread discussions about the future of technology and its societal implications. Despite its controversial release [61], the game's themes of technological advancement, corporate power, and social inequality resonate

with contemporary concerns. This cultural relevance makes the game a valuable lens through which to examine public perceptions and ethical considerations surrounding emerging technologies. The game's depiction of a dystopian future serves as a cautionary tale, emphasizing the need for careful consideration of the societal impacts of technological progress.

While *Cyberpunk 2077* draws on established themes from the cyberpunk genre, it offers a modern reinterpretation that incorporates contemporary technological trends and challenges. The game updates classic cyberpunk motifs – such as the fusion of human and machine, the omnipresence of corporate control, and the struggle for personal identity in a high-tech world – with new insights and scenarios that reflect the current state of technology. This blend of historical context and modern innovation provides fresh perspectives on familiar themes, making the game a relevant and insightful exploration of the future. Many of the technologies in *Cyberpunk 2077* have been previously explored in other cyberpunk works, but the game's interactive format and complex narrative structure provide a different medium for these explorations.

By comparing *Cyberpunk 2077* to its predecessors and contemporaries, we can identify what makes its approach unique. For example, its use of real-time decision-making and the impact of player choices on the storyline allow for a deeper engagement with the ethical dimensions of technological advancement. Compared to other sci-fi works like *Black Mirror*, *The Matrix*, and *Deus Ex*, *Cyberpunk 2077* offers a more intricate and participatory engagement with AI ethics, cybernetic augmentation, and dystopian futures. *Black Mirror* typically presents isolated cases of technological misuse, focusing on extreme hypothetical scenarios to highlight ethical dilemmas. While effective as a warning, its episodic nature means that it rarely examines the long-term systemic integration of AI or cybernetics into society. *The Matrix*, in contrast, deals with AI as a singular existential threat, portraying artificial intelligence as an antagonistic force that controls human existence rather than an embedded part of daily life. This perspective, while philosophically rich, does not explore the incremental and often ambiguous ways AI technologies shape modern societies. *Deus Ex*, as an interactive game, incorporates themes of cybernetic augmentation but does so within a structured gameplay framework where augmentation is primarily treated as a functional game mechanic rather than a societal force. While it raises ethical questions about human enhancement, it presents augmentation as an individualistic choice rather than an inevitable consequence of technological and corporate expansion. *Cyberpunk 2077*, by contrast, embeds cybernetic augmentation into the fabric of its world, illustrating the economic, social, and political forces that drive and regulate human enhancement. The game challenges players to consider augmentation beyond personal utility, instead emphasizing its societal consequences—such as corporate exploitation, inequality, and systemic dependency on technology. In contrast, *Cyberpunk 2077* offers a more nuanced examination of AI ethics. Rather than depicting AI as inherently benevolent or malevolent, the game presents artificial intelligence as a contested space of power struggles, corporate interests, and ethical ambiguity. AI entities in the game exist within a legal and economic framework, reflecting contemporary debates about AI governance, autonomy, and accountability, situating AI within a broader societal structure.

Cyberpunk 2077 allows open-ended exploration of corporate AI governance, bio-enhancement inequalities, and algorithmic bias through dynamic player interactions. By forcing players to confront the social consequences of these augmentations firsthand, rather than through passive observation, the game functions as an active foresight tool, immersing users in speculative futures they must navigate, rather than merely witness.

However, existing analyses often overlook the game's potential as a foresight tool that actively engages players in speculative futures, rather than merely presenting them. We examine how *Cyberpunk 2077* reflects and challenges current technological paradigms, offering a dynamic, interactive platform that bridges science fiction with real-world technological trends and societal implications.

2.5. Technologies in *Cyberpunk 2077*

The specific technologies presented in *Cyberpunk 2077* have also been the subject of academic inquiry, with researchers exploring the games representation of various technological domains. Human augmentation, for instance, has been a subject of interest in various studies, examining the ethical, social, and philosophical implications of integrating technology into the human body [62], [63], [64]. Similarly, brain-computer interfaces, as depicted in *Cyberpunk 2077*'s *Braindance* concept, have been investigated in terms of their potential for revolutionizing human cognition and communication [65].

Other areas of research related to the games themes include simulated reality [66], digital representation and information access [67], smart homes and appliances [68], [69], autonomous vehicles [70], and advanced smart assistants [71]. These studies provide insights into the current state of these technologies, their challenges, and potential future developments.

3. Methodology

In constructing a solid theoretical foundation for our analysis, several established theories and frameworks have been identified as particularly relevant. Scenario Planning [72] is a foresight tool widely used in both corporate foresight and academic studies, allowing researchers to explore multiple future scenarios based on current trends. This approach facilitates understanding the potential implications of different trajectories, helping organizations and policymakers prepare for diverse possibilities. Actor-Network Theory (ANT), developed by Latour and Callon [73], provides a framework for examining the interconnectedness of technology, human actors, and institutions. ANT is particularly useful in understanding how technologies shape and are shaped by social structures, emphasizing the complex dynamics between human and non-human actors. Critical Future Studies [74] further contribute to this foundation by focusing on the ethical implications and power dynamics in future-oriented thinking. This framework critiques dominant narratives about the future, advocating for more inclusive and pluralistic approaches to foresight. Lastly, Cyborg Theory, with key contributions such as Haraway's *A Cyborg Manifesto*, [75] explores the merging of human and machine, offering a critical perspective on technology's impact on human identity and the evolving boundaries between nature and technology.

Loosely based on these ideas, in this article, we aim to systematically analyze the key technological themes and advancements presented in *Cyberpunk 2077* and compare them with the current state of technology. To achieve this, we followed a three-step methodology, which we outline below.

3.1. Data collection

First, we collected data from the *Cyberpunk 2077* video game and its associated official materials, such as guides, trailers, and developer interviews. We focused on identifying the technological concepts and innovations present in the game, paying particular attention to the technologies that were emphasized in the presentation. This data served as our primary source for understanding the vision of the future as portrayed by the game developers.

The data collection included a detailed examination of the game content, focusing on the portrayal of various technologies and how they integrate into the game's narrative and gameplay. As a starting point, we dedicated approximately 200 h to replaying the game with a special emphasis on immersing ourselves in the technologies, taking notes of some key aspects presented to the player. For a more systematic data collection, we familiarized ourselves with a wide array of official guides and trailers. The guides provided in-depth insights into the game's world, offering detailed descriptions of the technological elements and their functionality within the game's universe. The trailers, on the other hand, served as a visual representation of these technologies, offering a glimpse into their potential real-world applications and implications.

A complementary source of our data collection was the analysis of developer and actor interviews. We reviewed a substantial corpus of over 10 interviews with the game's developers, sourced from various media outlets and gaming conferences, publically available on YouTube in popular gaming channels such as GameSpot, IGN, GamingBible, or GamesRadar. These interviews were instrumental in understanding the developers' perspectives on the future technologies they

envisioned and incorporated into the game. They provided valuable context regarding the rationale behind the inclusion of specific technologies and how they envisage these technologies evolving in the real world, and the actors approach towards the interaction with the world.

3.2. Thematic analysis

Next, we conducted a thematic analysis of the collected data to identify key themes and technological advancements within the game. We followed Braun and Clarkes six-phase approach to thematic analysis [76], [77], which involved:

- Familiarizing ourselves with the data by immersing ourselves in the game world and closely examining the identified technologies.
- Generating initial codes by identifying interesting features of the data and systematically coding these features.
- Searching for themes by collating the generated codes into potential themes.
- Reviewing themes by refining the identified themes and ensuring that they are supported by the data.
- Defining and naming themes by providing clear definitions and names for each theme.
- Producing the final report by presenting the findings of the analysis in a coherent manner.

This process allowed us to identify the key technological themes in Cyberpunk 2077 and organize them into coherent categories for further analysis.

3.3. Comparison with current technologies

In our final analytical phase, we compared the futuristic technological themes from Cyberpunk 2077 with the current state of technology, leveraging our expertise in AI and edge computing. Instead of a systematic review, we adopted a narrative style approach, using our expertise to select, interpret, and synthesize the relevant literature and recent technological developments. To conduct this expert-driven narrative review, we followed these steps:

Step 1. Keyword-Driven Literature Exploration: We chose relevant keywords to explore the existing literature. We used databases such as IEEE Xplore, ACM Digital Library, PubMed, and ScienceDirect, employing a combination of keywords related to each identified technology theme.

This search was complemented by the use of Google Scholar to ensure the inclusion of gray literature and the latest research outputs. This exploration was not limited to systematic database searches but extended to identifying seminal and cutting-edge papers indicative of current trends.

Step 2. Narrative Synthesis: We synthesized the findings from our literature exploration in a narrative format, highlighting how current developments in AI and edge computing align with or diverge from the technologies portrayed in Cyberpunk 2077.

Step 3. Expert Insight and Forward-Looking Perspectives: We studied the selected literature and suggested future works to speculate on probable next developments, potential roadblocks, and emerging opportunities. This included a critical examination of the feasibility, ethical considerations, and potential timeframes for the realization of the technologies depicted in the game.

Step 4. Comparative Analysis: We conducted a comparative analysis of the identified themes against the backdrop of current technologies, assessing their realism, potential impact, and how they reflect or diverge from existing technological trajectories.

4. Themes and technologies

In this section, we present the key themes and technologies identified through our thematic analysis of Cyberpunk 2077. These themes encompass the core aspects of the games portrayal of future technological advancements and provide a framework for understanding the games vision of the future. A summary of the findings, including the links between depicted future technologies, current state of the art and possible research directions is shown in Table 1. This table juxtaposes the futuristic technologies portrayed in Cyberpunk 2077 against the current technological landscape and outlines potential research directions that could bridge the gap between todays reality and the games

visionary future. Table 1, shows insights into the convergence and divergence between Cyberpunk 2077's fictional world and our present technological trajectory, with a speculation on the foresight and the plausible evolution of these technologies. The following sections, detail the connections between the tables columns and the relevant literature.

4.1. Overarching themes

In addition to the specific technologies and advancements, Cyberpunk 2077 presents several overarching themes that pervade its vision of the future. Reflecting on the broader trends and principles that shape the development and application of new technologies in the games world, these overarching themes provide a broader context for understanding the specific technologies and advancements present in Cyberpunk 2077, highlighting key principles and trends that shape the games portrayal of the future.

- Ubiquitous Sensing: (Everything Sensing) In the games future, virtually every device, object, and even human beings are integrated with a multitude of sensors. This constant sensing enables real-time data collection and analysis in a truly distributed manner, which in turn drives more informed decision-making and responsive technologies.

- Interconnectivity: (Everything Connected) The world of Cyberpunk 2077 is characterized by ubiquitous interconnectivity, with all devices, systems, and even humans constantly communicating and collaborating. This interconnectivity allows for the formation of very large-scale networks of sensors and data sources, enabling more efficient and coordinated responses to various challenges.

- Autonomy: (Everything Autonomous) Many technologies in the game are autonomous, meaning they can operate independently without direct human intervention. This autonomy extends to various devices and systems, ranging from computing resources and communication networks, to vehicles and robots.

- Multimodality: (Everything Multimodal) Technologies in Cyberpunk 2077 often embrace multimodality, capturing and processing various types of sensory data simultaneously.

This multimodal approach enables more comprehensive and nuanced understanding and analysis of complex situations.

- Personalization: (Everything Personalized) A strong emphasis is placed on the personalization of the games world, with technologies adapting to individual users locations, needs, and preferences. This personalization allows for more tailored and effective solutions to various challenges and enhances overall user experiences, allowing people to effectively experience the world in a truly unique manner.

- Bio-Integration: (Everything Bio-Integrated) The games vision of the future also includes the convergence of the physical, artificial, and biological worlds. Technologies are seamlessly integrated into human bodies and natural environments, creating new hybrid systems that combine the strengths of each domain.

- Digital Security: (Everything Secure) The world of Cyberpunk 2077 shows unwavering dedication to securing all digital interactions, from personal communication to vital infrastructure. In this hyper-connected society, where everything is linked digitally, constant readiness against cyber threats is essential. The narrative highlights the need for robust safeguards and vigilance against potential breaches, in a way that intertwines with human abilities.

The main technologies of the game have been grouped in six different thematic groups: Human Augmentation, Brain Computer Interfaces, Digital Representation and Information Access, Smart Environments and Personalization, Autonomous Vehicles and Transportations and Self-aware AI and Assistants.

4.2. Human augmentation and cybernetic enhancements

Cyberpunk 2077 envisions a future where humans are augmented through biotechnology, cybernetic implants, and genetic modifications. However, in this envisioned society, not all individuals opt for or can afford these enhancements, revealing disparities and philosophical tensions among its denizens. In Cyberpunk 2077, human augmentation is not just a technological

backdrop but a central element of the narrative. Characters like V, the protagonist, and Johnny Silverhand, a digital ghost of a historical character, exemplify the profound personal and societal impacts of cybernetic enhancements. For instance, V's journey through the game is deeply intertwined with their reliance on cyberware, which raises questions about identity, humanity, and the ethical boundaries of human enhancement. These narrative elements provide a nuanced view of the potential and pitfalls of such technologies, setting *Cyberpunk 2077* apart from other generic portrayals of cybernetic augmentation in the cyberpunk genre. This theme includes technologies such as cybernetic prosthetics, artificial biological augmentations, and genetic enhancements that expand human capabilities beyond their natural limits. Fig. 2 shows examples of human augmentation technologies as depicted in *Cyberpunk 2077*, underscoring the game's portrayal of cybernetic enhancements as status symbols highlights how technological advancements can reinforce social hierarchies, and portraying human augmentation as both a tool for empowerment and a marker of socio-economic divides.

Cybernetic Prosthetics replace or enhance body parts with advanced mechanical or electronic components, providing users with improved strength, agility, or sensory perception [78]. In *Cyberpunk 2077*, some examples include robotic limbs, advanced ocular implants, and neural interfaces. Today, prosthetic limbs have evolved significantly, incorporating advanced materials, sensors, and electronics to provide improved functionality. Some prosthetics can be controlled by the user's muscle signals or through brain-computer interfaces, while others offer sensory feedback [78], [79]. However, these prosthetics are still limited in terms of sensory perception, dexterity, and overall performance compared to the advanced prosthetics portrayed in *Cyberpunk 2077*. To reach the level of cybernetic prosthetics in the game, research should focus on developing more sophisticated hardware, software, and signal processing techniques that can better interpret user intent and accurately mimic natural movements [80]. Additionally, integrating advanced sensors, materials, and power systems will be crucial for creating more realistic and high-performance prosthetics [121], [122]. Academic and industrial efforts, such as Open Bionics Hero Arm, MIT Media Labs Biomechatronics Group, and Johns Hopkins University's Modular Prosthetic Limb demonstrate significant advancements in bionic limbs with sensory feedback and neural integration. Recent advances in haptic feedback in robot-assisted surgery, emphasize how tactile interfaces improve precision and patient outcomes [94]. As the trust users place in their cybernetic enhancements hinges on the guarantee that they are secure from external manipulations, future research should also prioritize establishing robust security protocols and standards.

A representation of human augmentation technologies in *Cyberpunk 2077* helps showcasing their role as status symbols and the potential socio-ethical implications of uneven access.

Artificial Biological Augmentations involve the incorporation of synthetic or lab-grown biological components into the human body, such as replacement organs, synthetic muscles, or artificial blood. These augmentations can improve health, vitality, or physical performance. In the present time, advances in regenerative medicine and tissue engineering have led to the development of lab-grown organs and tissues, such as skin, blood vessels, and cartilage, with industrial and academic initiatives such as Organovos bioprinting [47], Wake Forest Institute's regenerative organ research, and the University of Minnesota's 3D-printed heart models, which showcase cutting-edge work in artificial biological augmentations. Currently, the ethical implications of using human augmentation are still carefully considered, ensuring they are used for genuine therapeutic purposes and not merely for enhancing one's natural abilities beyond ethical boundaries [81]. However, these technologies are still in the early stages of development, and widespread clinical application remains limited. To achieve the level of artificial biological augmentations seen in *Cyberpunk 2077*, researchers need to continue exploring methods for generating functional, lab-grown tissues and organs that can seamlessly integrate with the human body. This includes the development of advanced biomaterials, cell sources, and fabrication techniques, as well as overcoming challenges related to vascularization, innervation, and immune compatibility [81], [82].

Genetic Modifications in *Cyberpunk 2077* entail the alteration of an individual's genetic code to enhance physical or mental traits, such as increased intelligence, physical strength, or resistance to diseases. To date, genome editing technologies such as CRISPR/Cas9 have facilitated gene modification with remarkable precision, thereby contributing to significant advancements in both gene therapy and the creation of genetically modified organisms [83], with private initiatives such as Intellia Therapeutics highlighting the efforts. Despite these advancements, a comprehensive understanding of the genetic influences on human traits and diseases remains elusive [123]. Recent scholarly reviews indicate that the complexities of genetic impact on specific attributes, such as sports performance, have yet to be fully elucidated, suggesting that similar limitations may apply to the enhancement of other human traits [84], [85]. To achieve the level of genetic modifications depicted in the game, researchers must continue to refine genome editing techniques to increase their accuracy, efficiency, and safety. This includes developing new gene editing tools, delivery methods, and strategies for minimizing off-target effects. Additionally, addressing ethical, legal, and societal implications of human enhancement through genetic modifications will be crucial for their responsible implementation.

4.3. Brain-computer interfaces and simulated reality

The game introduces the concept of *braindance*, an advanced recording technology capable of capturing and replaying sensory input and external sensor data. This innovation enables users to immerse themselves in past experiences on various sensory levels. *Braindance*, a technology that allows users to experience recorded memories and sensations. This narrative device is used to explore themes of escapism, privacy, and the commodification of experiences. The missions involving *Braindance* technology provide a critical commentary on how immersive experiences can be manipulated and controlled, offering a distinctive perspective that enriches the broader discourse on virtual reality. This theme also encompasses significant advancements in simulated reality, facilitating the creation of exceptionally realistic virtual environments suitable for diverse purposes, including training, education, and entertainment. Within the *Cyberpunk 2077* universe, certain inhabitants are deeply immersed in this hyper-connected environment, often prioritizing their experiences through such interfaces. However, this heightened reliance on technology has given rise to mental health concerns, notably the condition known as *cyberpsychosis*. Fig. 3 provides visual examples of Brain-Computer Interface (BCI) technology as portrayed in *Cyberpunk 2077*, where their rise introduces new considerations for mental health, and represent both the transformative potential of neural integration and the ethical challenges of privacy and security.

Brain-Computer Interfaces and haptics (BCIs) play a critical role in *Cyberpunk 2077*, depicted as devices facilitating direct communication between the human brain and external devices, such as neuralware and *braindance* systems. Modern BCIs have made significant progress in recent years, with applications ranging from assisting individuals with disabilities to controlling virtual environments [124]. However, current BCIs are primarily non-invasive and have limitations in terms of signal quality, bandwidth, and user experience compared to the highly advanced BCIs depicted in the game [86]. Moreover, as these interfaces access and transmit neural data, concerns about data security and unauthorized access emerge, and need cybersecurity measures to safeguard users' neural and personal information [88]. To advance the state of brain-computer interfaces (BCIs) to the level depicted in *Cyberpunk 2077*, targeted research is needed to improve their connectivity, miniaturization, and biocompatibility. This involves work on minimally-invasive methods, faster data transmission, new materials, improved signal storage, and better algorithms for signal processing [87]. Such advancements in BCIs are also considered crucial for their application in future manned space missions [89], an aspect highlighted in *Cyberpunk 2077*. Neuralinks implantable BCIs, Metas CTRL-Labs wrist-worn neural interfaces, and DARPA's N3 program illustrate groundbreaking progress in brain-computer interface technologies [125]. Recent advancements in haptic technology complement the portrayal of BCI. Research in hand-based haptic interaction in virtual reality, has highlighted how tactile feedback enhances immersion and realism in digital environments [92], suggesting that haptic interfaces could bridge the gap

between the virtual and physical worlds. Haptics are already revealing their potential to reshape societal interactions, particularly in healthcare and entertainment [93].

An example of brain–computer interfaces (BCIs) in *Cyberpunk 2077*, helps reflecting on the potential of BCIs for education and cognitive enhancement, as well as the ethical challenges in ensuring accessibility and mental health safety.

Simulated Reality in *Cyberpunk 2077* features highly immersive simulated reality experiences, including virtual reality (VR) and augmented reality (AR) systems that can seamlessly blend digital content with the physical world. VR and AR technologies have seen rapid advancements in recent years, with applications in gaming, education, and training [126], [127]. Meta Quest Pro, Magic Leap AR devices, and Stanford's Virtual Human Interaction Lab exemplify advances in immersive VR/AR systems and their real-world applications. However, these systems still have limitations in terms of display resolution, field of view, induction of motion sickness, and user interaction compared to the near-perfect simulated reality portrayed in *Cyberpunk 2077* [90], [91]. However, to reach the level of simulated reality seen in the game, research should focus on developing AI-generated, procedural content, improving display technologies, and creating realistic interactions within virtual environments. This includes advancing the development of haptic feedback, gesture recognition, and environmental sensing technologies, as well as optimizing software and hardware for more immersive experiences [95], [96].

4.4. Digital representation and information access

Cyberpunk 2077 presents a future where digital information is seamlessly integrated into the physical world through standardized protocols and real-time authentication based on biometrics. Amidst these advancements, unlawful access to information emerges as the most prevalent act of theft, highlighting the perpetual game between security and intrusion. This theme includes the development of digital twins, advanced biometric access systems, and improved information transfer and security. In *Cyberpunk 2077*, digital representation and information access are vividly portrayed through the concept of the Net, a vast cyberspace that exists entirely in the digital realm. This virtual world is inhabited by AI constructs and accessed by netrunners—hackers who can jack into cyberspace to gather information or manipulate digital systems. The mission *I Walk the Line* showcases this aspect of the game, as V ventures into cyberspace to confront the rogue AI, Brigitte. This mission illustrates the complex interactions within the Net, highlighting how information is accessed, controlled, and defended in a purely digital environment. Fig. 4 shows examples of the Cyberspace, the Digital Representation of the world as depicted in *Cyberpunk 2077*, illustrating the potential applications of digital twins in predictive modeling and real-time simulations and how the technology has heavy dependences on the ability to generate realistic interactions.

Digital Twins are envisioned in the world of *Cyberpunk 2077* in a way where virtual replicas of physical objects or systems are commonplace, and show ubiquitous, and instantaneous capabilities. They allow for enhanced simulation, monitoring, and optimization of real-world assets. Digital twin technology is already being used in various industries, such as manufacturing, infrastructure, and aerospace, for performance monitoring, predictive maintenance, and decision-making support [97]. Siemens and IBM's manufacturing twins, GE Digital's energy sector applications, and numerous European initiatives covered by the Horizon Europe projects showcase the practical use of digital twin technologies. However, the level of integration and sophistication depicted in *Cyberpunk 2077* is still beyond the capabilities of current technology. To achieve this level of digital twin integration, research should focus on developing standardized protocols for data exchange, improving real-time data processing, and enhancing the scalability and interoperability of digital twin systems. This includes the development of advanced machine learning and AI techniques for data analysis, optimization, and prediction. In addition, research is needed into integrating biometric and biological data into human digital twins that can populate the virtual replicas of the physical world [98].

Digital representation of the world as depicted in Cyberpunk 2077, offering insights into the potential of real-time simulations and predictive modeling of digital twins.

Advanced Biometric Authentication systems provide the Cyberpunk 2077 works with secure, personalized access to various services and facilities based on unique physical characteristics, such as fingerprints, iris patterns, or facial features. Currently, biometric authentication is widely used in modern security systems and personal devices, with technologies such as fingerprint scanners, facial recognition, and voice recognition becoming increasingly common [99]. Apples Face ID, NECs Bio-IDiom, and Microsoft Azures multimodal biometrics represent recent commercial success in reliable biometric authentication. However, the game presents a future where biometrics are globalized, more pervasive and integrated into everyday life. To reach the level of biometric authentication seen in Cyberpunk 2077, research should focus on improving the accuracy, reliability, and security of biometric systems, especially when facing adversarial attacks [101]. In addition, future research efforts should prioritize the development of continuous authentication methodologies, ensuring that users are continuously verified throughout their interactions with systems, enhancing both security and user experience [100]. This includes developing new sensors, algorithms, and privacy-preserving techniques to ensure the robustness and user-friendliness of the solutions.

Information Transfer and Security mechanisms, allowing for rapid and secure exchange of data across various networks and devices of the Cyberpunk 2077 universe. In recent times, modern wireless communication networks (5G/6G) have made significant strides in terms of speed, reliability, and security, and future architectures plan to increase extend their capabilities and flexibility beyond what is usable now [45]. However, they are not yet seamlessly integrated and foolproof. To achieve the level of information transfer and security presented in the game, research should focus on developing new communication protocols, encryption algorithms, and network architectures that can support the rapid, secure, and seamless exchange of data. This includes the exploration of semantic communication, edge computing, and advanced cybersecurity techniques to protect against potential threats and vulnerabilities [102].

4.5. Smart environments and personalization

The game envisions a future where homes and appliances are smart, interconnected, and highly personalized. Cyberpunk 2077 showcases smart environments through various in-game settings that respond dynamically to the players actions and preferences. The luxurious apartments and corporate offices in Night City are equipped with advanced smart home technologies, including personalized lighting, climate control, and interactive digital assistants. For example, V's apartment features a smart mirror that provides personalized health updates and news. The mission The Hunt takes players through a high-tech, personalized living space designed for a child, complete with responsive toys and educational tools that adapt to the child's learning pace and interests. These smart environments illustrate the potential of personalized technology to enhance daily living while also prompting reflections on privacy, data security, and the ethical use of personal information. Technologies in this theme include different home appliances such as smart mirrors, fashion applications, smart fridges, and various home automation systems that adapt to individuals needs and preferences. Fig. 5 shows examples of different smart environments and personalization technologies present in Cyberpunk 2077, highlight the potential for adaptive systems to enhance user experiences while also raising concerns about data privacy and surveillance.

The integration of AI-driven personalization in Cyberpunk 2077 mirrors current advancements in education, advertising, and design. AI is showing transformative potential in tailoring educational experiences to individual learning styles, a concept echoed in the game's adaptive environments [128]. Similarly, AI personalized customer experiences showcase its utility in commercial applications [108], while AI-assisted design, emphasizes how personalization fosters creativity in complex processes [105]. These examples demonstrate how AI is already increasingly embedded in various industries.

Personalized smart environments in Cyberpunk 2077, emphasizing the usefulness of adaptive systems and personalized user experiences.

Smart Mirrors and Fashion Applications can analyze a users appearance and suggest fashion choices or cosmetic adjustments, enabling users to change their clothing or appearance virtually. Even nowadays, smart mirrors with limited capabilities, such as displaying weather or news updates, already exist [103]. MemoMis Memory Mirror, Sephoras Virtual Artist, and Amazons Echo Look highlight innovative applications of AR in fashion and personal styling. Current technology is progressing to offer detailed fashion advice or enable virtual clothing changes, but challenges remain in how to input commands in real-time with great accuracy [129]. To achieve the level of smart mirrors and fashion applications seen in the game, research should focus on improving computer vision, image recognition, and AR/VR technology. This includes developing algorithms for real-time clothing and appearance analysis, virtual clothing manipulation, but specially, seamless AR/VR integration [104].

Smart Appliances and Home Automation systems that coordinate their behavior, learn from users preferences, and adapt to individual needs are omnipresent in Cyberpunk 2077. Nowadays, smart appliances and home automation systems are already available, offering a range of capabilities such as remote control, energy management, and simple learning algorithms. Google Nest, Samsung SmartThings, and Philips Hue showcase how smart appliances and home automation are becoming seamlessly integrated into daily life. However, current systems generally have limited coordination and personalization capabilities compared to those portrayed in the game [106]. To achieve the level of smart appliances and home automation seen in Cyberpunk 2077, research should focus on improving inter-device communication, machine learning algorithms, and user behavior analysis. This includes developing standards and protocols for seamless device integration, advanced algorithms for learning user preferences and predicting needs, and improved techniques for analyzing user behavior and intentions. Research into coordination and personalization of devices with a complex socio-material perspective is paramount [107].

4.6. Autonomous vehicles and transportation

Fully autonomous vehicles, including cars and drones, are a key feature of Cyberpunk 2077s futuristic setting, where they often coexist with regular ones. These smart vehicles, rely on advanced sensor and communication systems to navigate complex urban environments and interact with unpredictable elements, such as pedestrians and other vehicles. Flying drones are ubiquitous in Night City, performing tasks ranging from surveillance to delivery, reflecting the seamless integration of autonomous systems into various aspects of urban life. Autonomous vehicles and advanced transportation systems are integral to the games depiction of a futuristic urban environment. Players frequently interact with autonomous driving cars, which offer a glimpse into the potential of self-driving technology. Missions like Human Nature and Tune Up explore the intricacies of these technologies, showcasing their role in everyday life and their potential vulnerabilities. Fig. 6 shows flying cars, personal drones and autonomous driving and parking technology as seen in Cyberpunk 2077, exemplifying the balance between innovation and the challenges of urban infrastructure in a highly connected world.

Autonomous Ground Vehicles are showcased in the game as driverless cars and taxis capable of traversing intricate urban settings with minimal human intervention, in a coordinated manner. In the current state of the art, autonomous ground vehicles have made significant progress in recent years, with companies such as (e.g.) Tesla, Waymo, and Cruise developing and testing self-driving cars. However, current autonomous vehicles still require human oversight and are not as advanced and fully autonomous as those depicted in the game [109]. To reach the level of autonomy portrayed in Cyberpunk 2077, research should focus on improving simultaneous localization and mapping (SLAM), sensor fusion, and decision-making algorithms for vehicles operating in dynamic and complex environments. Additionally, the development of standardized communication protocols and infrastructure to support vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication will be crucial in facilitating safe and efficient autonomous transportation.

Improvements of self-driving models to handle out-of-distribution examples and fringe cases need to be greatly improved [110].

Autonomous vehicles and transportation in Cyberpunk 2077, underscoring the interplay between innovation and urban infrastructure.

Autonomous Flying Vehicles such as drones and personal aircraft, are used in the Cyberpunk 2077 universe for transportation, surveillance, and delivery purposes. While autonomous drones are currently available for various applications, including aerial photography, mapping, and inspection, fully autonomous personal aircraft are still in the early stages of development and testing, while EHangs passenger drones, NASA's Advanced Air Mobility project, and Joby Aviation's eVTOL developments are leading the advancements in autonomous flying vehicles. To achieve the level of autonomous flying vehicles seen in the game, research should focus on enhancing flight control systems, collision avoidance, and air traffic management for operation in urban environments. This includes developing advanced computer vision for sensing and understanding [111], AI algorithms for path planning, real-time decision-making, and robust communication systems to facilitate seamless interaction with other aerial vehicles and ground-based infrastructure [112].

Autonomous Delivery Robots in Cyberpunk 2077 are small ground-based delivery robots that navigate sidewalks and streets to transport packages and goods. Companies such as Starship Technologies, Nuro, and Amazon have developed and tested autonomous delivery robots, but their deployment is limited and not yet widespread and ubiquitous [113]. To achieve the level of autonomous delivery robots seen in the game, research should focus on improving navigation algorithms, obstacle detection and avoidance, and human-robot interaction to ensure the safe and efficient operation of these robots in complex urban environments. Autonomous robot-driven delivery systems should address realistic constraints, incorporate uncertainty through stochastic modeling, and enable dynamic routing through multi-objective optimization. Additionally, the development of standardized communication protocols and infrastructure to support robot-to-robot (R2R) and robot-to-infrastructure (R2I) communication will be vital in facilitating their large-scale deployment [114].

4.7. Advanced artificial intelligence and smart assistants

The game showcases advanced AI systems that are self-aware and capable of personalized interactions based on users' traits, emotions, and intentions. These smart assistants provide users with context-aware support and adapt their behavior to better serve individual needs. Cyberpunk 2077 explores the ethical and existential implications of artificial intelligence through characters like Delamain, an AI-driven taxi service that grapples with self-awareness and autonomy. This subplot not only highlights the technical challenges associated with AI but also delves into the moral and ethical considerations of creating self-aware machines. By engaging players in these scenarios, Cyberpunk 2077 offers a unique interactive medium to explore these complex issues in a way that traditional media cannot. Fig. 7 shows natural interaction with AI and assistants, illustrating the depiction of self-aware AI in Cyberpunk 2077 which aims at raising profound ethical questions about consciousness, rights, and societal integration.

Self-aware AI systems are portrayed in Cyberpunk 2077 as being capable of complex reasoning, decision-making, and natural interactions with humans. While AI has made significant progress in recent years, current AI systems are generally narrow in their capabilities and are not yet fully self-aware or capable of general intelligence [115]. In the very recent times, Large Language Models (LLM) have shown incredible capabilities beyond what was previously possible, and the rapid pace of development is only expected to bring new possibilities at a fast pace. However, in order to achieve the level of self-aware AI seen in the game, research should focus on developing algorithms and architectures that enable AI systems to reason about their own existence, understand their limitations, and adapt to new situations. This includes advancing areas such as artificial general intelligence (AGI), machine learning, and cognitive computing, while taking enough time to discuss ethical topics that include AI alignment, and bias awareness [116]. The anthropomorphic

portrayal of AI agents in *Cyberpunk 2077* reflects ongoing research into human-AI interaction. Human-like features influence user trust and engagement with AI agents, suggesting that human-like AI interactions can foster emotional connections [130], while the rise of large language model-based agents, can automate further communication and decision-making systems [117]. These developments reinforce the narrative complexities explored in the game, where AI agents balance autonomy and ethical considerations.

Self-aware artificial intelligence as shown in *Cyberpunk 2077*, provoking questions about consciousness, ethics, and societal integration.

AI-based Personal Assistants are featured in *Cyberpunk* as agents that are able to predict user needs, recognize emotions, and adapt their behavior accordingly. Currently, AI-based personal assistants like Siri, Alexa, and Google Assistant have become increasingly popular, but their capabilities are limited compared to those portrayed in the game. While the most widespread smart assistants used to rely on predefined responses and were not able to respond to a wide variety of scenarios in a credible manner, recent advances in LLMs, including fine-tuning of the models for human conversation and Reinforcement Learning from Human Feedback, have pushed the field to new heights, increasing the usability of the systems almost exponentially. However, current assistants, although able to influence human cognition, are still not able to fully understand or adapt to users emotions and intentions [118]. To achieve the level of AI-based personal assistants seen in the game, research should focus on improving not only natural language processing, but emotion recognition, and context-aware reasoning for chain of thoughts [119]. This includes developing advanced algorithms for understanding and generating natural language from multimodal data [131], but most importantly detecting and interpreting human emotions, and predicting user needs based on context and history [120].

5. Discussion

Our analysis of *Cyberpunk 2077*'s vision of the future reveals a range of themes and technologies with the potential to significantly impact society, the economy, and culture. While some technologies portrayed in the game, like AI and brain-computer interfaces, are rapidly evolving, others remain speculative, highlighting the dual nature of technological progress—its tangible advancements and its aspirational horizons. Unlike traditional foresight methods, *Cyberpunk 2077* serves as an active, participatory foresight tool where players engage directly with speculative futures, experiencing the socio-ethical implications of advanced technologies firsthand. This aligns with Scenario Planning and Design Fiction approaches by creating immersive, real-time scenarios that allow players to explore complex technological futures in ways that extend beyond conventional narrative formats [22], [23]. This interactive approach can be adapted (e.g.) in foresight workshops, allowing participants to engage directly with speculative futures, enhancing their ability to critically assess the socio-ethical implications of emerging technologies. One notable implication of the widespread adoption of human augmentation technologies is the potential for exacerbating existing social inequalities [132]. Table 1 showed how current research directions should focus on improving hardware, software and sensor technologies. Access to these enhancements may be limited by factors such as cost, availability, and social acceptance, potentially leading to a divide between augmented and non-augmented individuals. This divide may manifest in terms of employment opportunities, social mobility, and even basic human rights. *Cyberpunk 2077* vividly portrays a future where advanced technologies are unevenly distributed, exacerbating existing social inequalities. The game's depiction of Night City, with its stark contrast between the affluent, technologically augmented elite and the impoverished, technology-deprived underclass, serves as a powerful critique of current and potential future socio-economic disparities. This narrative allows us to explore the implications of a society where access to life-enhancing technologies is determined by wealth and social status. Night City, as depicted in *Cyberpunk 2077* (Fig. 1), serves as a visual representation of how technological progress can exacerbate socio-

economic disparities. The stark contrast between its affluent, augmented elite and impoverished underclass highlights the risks of unequal technological access.

For instance, the game's portrayal of cybernetic enhancements as status symbols highlights how technological advancements can reinforce social hierarchies. This scenario suggests the importance of developing policies and frameworks to ensure equitable access to emerging technologies, mitigating the risk of deepening social inequalities. Additionally, the dominance of mega-corporations in controlling technological access and innovation, as depicted in the game, emphasizes an emerging need for regulatory frameworks to ensure equitable distribution and control of emerging technologies.

While themes of AI ethics, cybernetic augmentation, and dystopian futures have been widely explored in science fiction, *Cyberpunk 2077*'s approach differs by embedding these concerns into a complex socio-economic framework rather than presenting them as isolated moral dilemmas. Unlike *Black Mirror*, which often portrays AI and cybernetic risks as extreme, episodic cautionary tales, *Cyberpunk 2077* situates these technologies within ongoing power struggles, highlighting their long-term systemic consequences. Similarly, while *The Matrix* envisions AI as an external existential threat, *Cyberpunk 2077* reflects contemporary concerns by depicting AI as an integrated but exploitative force within human society. Compared to *Deus Ex*, which focuses on augmentation as a personal ethical choice, *Cyberpunk 2077* illustrates augmentation as an economic necessity shaped by corporate interests, reinforcing existing inequalities rather than simply offering individual empowerment. These distinctions emphasize the importance of analyzing technological advancements not only as ethical dilemmas but also as broader structural forces that shape social and economic realities.

The themes explored in *Cyberpunk 2077* offer valuable insights for practical applications across multiple fields. In education, technologies like brain-computer interfaces and immersive simulations could revolutionize learning through personalized and adaptive environments, enabling real-time feedback and experiential training. For policymakers, the game underscores the need for equitable access to emerging technologies, highlighting the importance of proactive regulations to address disparities, privacy concerns, and ethical dilemmas surrounding AI and human augmentation. Similarly, technology developers can draw from the interconnected ecosystems depicted in the game to prioritize scalable, secure, and inclusive innovations, fostering trust and minimizing risks in increasingly autonomous systems. By addressing these challenges, the lessons from *Cyberpunk 2077* can help shape more equitable and sustainable technological futures.

The rise of brain-computer interfaces and immersive simulated realities introduces new considerations for mental health [133]. Prolonged immersion might blur the lines between real and virtual experiences, leading to cognitive overload, addiction, or challenges in distinguishing reality from the virtual. This suggests potential cognitive and psychological effects of immersive technologies, which imply a need of further research into their long-term impact on human perception and mental health. The description of technologies suggest heavy dependences on our ability to generate realistic interactions. If successful, educational systems and cognitive development patterns might also evolve in unforeseen ways as instant information access becomes commonplace.

As AI approaches higher levels of self-awareness and personalized interaction, ethical considerations about AI consciousness emerge [134]. These self-aware AIs could be treated as mere tools or as entities deserving of rights and recognition. Furthermore, the challenge of AI alignment – ensuring that the AIs' goals and behaviors align with human intentions and values – becomes increasingly crucial [135]. As AI systems become more complex and autonomous, there is a rising urgency to ensure they act in ways beneficial to humanity and avoid unintended consequences. Research directions suggest that ethical aspects, and the interpretation of subtle human emotions will have the biggest impacts in the successful development of the technology. These choices have profound implications for AI-human relationships, leading to shifts in societal norms and the potential for new dynamics in how humans relate to technology. The game's narrative raises critical questions about the nature of consciousness and the responsibilities of creators towards their

creations. By exploring these themes, we can derive original insights into the potential legal and ethical frameworks required to address the rights and welfare of self-aware AI, ensuring that their integration into society is handled with due consideration of their autonomy and moral status. Cyberpunk 2077 challenges the notion of Technological Determinism by presenting a fragmented future where technological adoption is uneven and often detrimental, reflecting Critical Future Studies emphasis on the ethical and power dynamics inherent in technological progress [28], [74]. The games portrayal of uneven technological uptake critiques the idea that technology inherently leads to societal improvement, underscoring the need for a more nuanced understanding of how innovations impact different communities. The frequent technological malfunctions shown in the game, reflect on our current experiences with emerging technologies. The frequently shown glitches in cyberware or failed AI, highlight the unpredictable and often detrimental impacts of over-reliance on technology, reinforcing the argument that technological progress is not inherently linear or beneficial.

The increasing interconnectivity and autonomy of technologies also raise concerns about privacy, security, and control. As devices and systems become more capable of sharing and processing data, the potential for misuse or abuse of this information grows [136]. The pervasive influence of mega-corporations in Night City reflects the games critical stance on corporate control over technology. Through storylines involving corporations like Arasaka and Militech, Cyberpunk 2077 critiques the monopolization of technological advancements and the socio-political ramifications of such power. These narratives serve to contextualize the games technological themes within a framework of corporate ethics and social justice, making it a valuable contribution to discussions on the future of technology and society. Ensuring robust security and privacy protections will be crucial in maintaining trust and minimizing the risks associated with these advancements. As suggested in Table 1, the development of advanced inter-device communication, secure protocols and semantic communication will impact the trust that we will place in these systems. Cyberpunk 2077 illustrates the potential for enhanced communication and efficiency but also exposes the risks of cyber-attacks and data breaches in such an interconnected environment. Our analysis extends this depiction to propose that the increasing interconnectivity of future technologies necessitates robust cybersecurity measures and a rethinking of privacy norms. The games scenarios involving hacking and data theft emphasize the importance of developing resilient security protocols and ethical guidelines for data management. By drawing from these narratives, we can contribute to the ongoing discourse on cybersecurity and privacy in the digital age, providing novel insights into the challenges and solutions for safeguarding interconnected systems.

Environmental concerns also arise, particularly when contemplating the sustainability of producing and maintaining these high-tech devices [137], [138]. All research directions suggest the need for multiple devices, resulting in an ultradensification of the technology and communications. Sustainable practices in both production and disposal phases of augmentation technologies become paramount to mitigate environmental impacts.

Cyberpunk 2077s exploration of human-machine integration aligns with Posthumanism and Cyborg Theory, particularly in its portrayal of characters who embody the merging of organic and artificial elements. This directly relates to the work of Haraway and Hayles, illustrating the lived experience of posthumanism in everyday life and challenging existing paradigms of human identity and autonomy [75]. The games complex portrayal of cyborgs as both empowered and marginalized reflects the broader societal debates around human enhancement. The game shows characters, who are almost entirely mechanical, blurring the boundaries between human and machine, challenging traditional notions of identity and agency in a posthuman society.

The focus on personalization and bio-integration in Cyberpunk 2077s world highlights the need for ethical considerations in the development and application of new technologies [139]. Throughout the game, players encounter various scenarios that force them to confront the ethical and social implications of emerging technologies. For example, the side quest involving the kidnapping of children for illegal brainwashing experiments highlights the dark side of technological misuse. Such missions underscore the games commitment to addressing techno-moral themes in a manner that is

deeply integrated into its storytelling and gameplay mechanics. Balancing the benefits of customization and individualized solutions with concerns about surveillance, manipulation, and loss of autonomy will be an ongoing challenge [140]. Our future ability to develop advanced biometric authentication with standardized protocols will impact our trust in the technology.

Cyberpunk 2077 challenges the notion of Technological Determinism by presenting a fragmented future where technological adoption is uneven and often detrimental, reflecting Critical Future Studies emphasis on the ethical and power dynamics inherent in technological progress [28], [74]. The games portrayal of uneven technological uptake critiques the idea that technology inherently leads to societal improvement, underscoring the need for a more nuanced understanding of how innovations impact different communities. The diverse city districts shown in the game, each one presenting a very marked own cultural identity with its technological adaptation, reflect a break from the universalist approach often seen in traditional foresight, emphasizing the need for inclusive and culturally aware technological planning. A key insight from our analysis is the importance of designing technologies that can coexist with existing preferences and systems. A large proportion of the research directions suggested assume a fast-pace development that might challenge our current practices. However, and despite the rapid pace of technological change, it is important to recognize that not all individuals, organizations or societies will adopt or embrace new innovations uniformly [141]. Reflecting on the process on how advanced technologies intersect with and influence spiritual beliefs and practices is also an interesting, largely unexplored topic [142]. In this context, designing technologies that can accommodate a range of preferences, abilities, and cultural contexts will help ensure more equitable and inclusive outcomes [143].

6. Conclusion

In this paper, we have explored the themes and technologies presented in the video game Cyberpunk 2077 as a lens to envision potential future technological advancements. Our thematic analysis, grounded in the juxtaposition presented in Table 1, and developed in the following sections, illustrated the value of science fiction and video games as tools for stimulating imagination and fostering critical discussion about the direction and implications of emerging technologies.

While the games portrayal of the future is undoubtedly speculative and stylized, it offers valuable insights into the opportunities and challenges associated with various technological advancements. By examining these themes and technologies, we can better anticipate the potential impact of new innovations on society, the economy, and culture and inform the development of technologies that are more inclusive, ethical, and adaptable to a diverse range of needs and preferences.

Ultimately, the study of science fiction and video games like Cyberpunk 2077 can serve as a catalyst for interdisciplinary dialogue, collaboration, and exploration, helping to shape a more informed and responsible approach to the development and application of future technologies.

CRedit authorship contribution statement

Miguel Bordallo López: Writing – review & editing, Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Conceptualization. Constantino Álvarez Casado: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI GPT-4 in order to improve the readability of the article and generate proper LaTeX code for the references, tables and subsections. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.